Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where
- $F(x)$ = Residual Mapping
- x = Identity Mapping

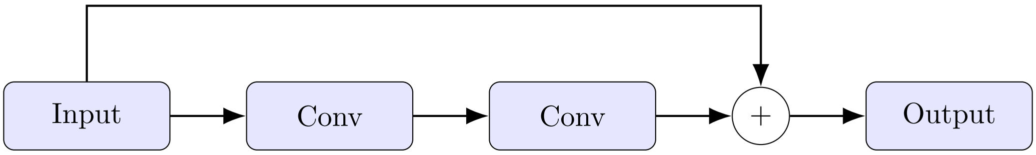


Figure 1: Residual Block in ResNet

Advantages
- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications
- Semantic segmentation
- Medical image analysis

- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

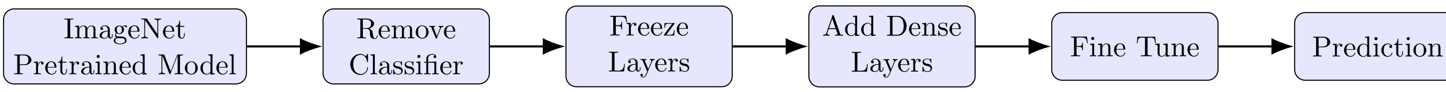


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.

4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

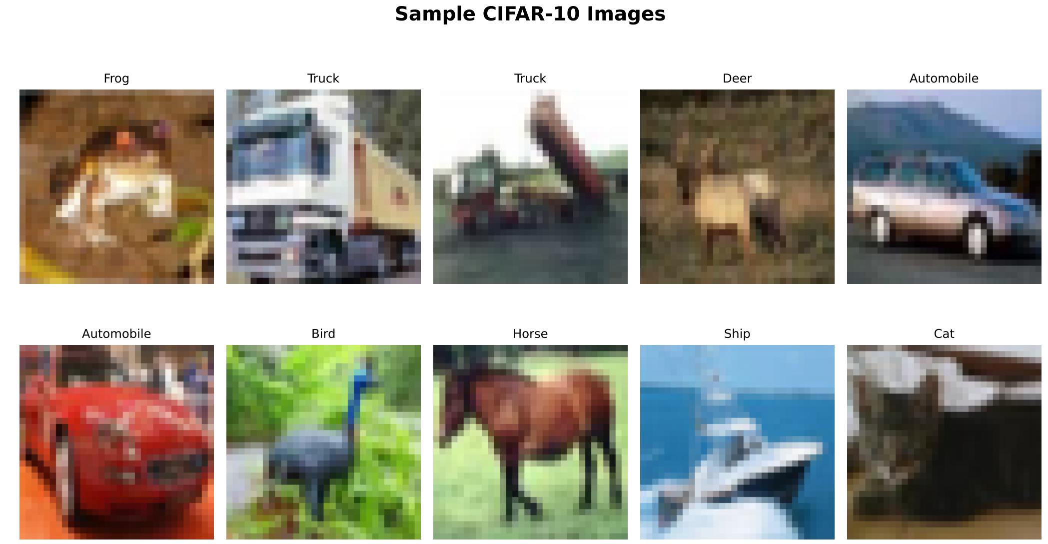
Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Source Code

The complete source code, datasets, generated plots, and report files for this experiment are available in the following GitHub repository:
GitHub Repository:
`https://github.com/VIKASNI-S/Deep-Learning-Lab/tree/main/Lab-4`

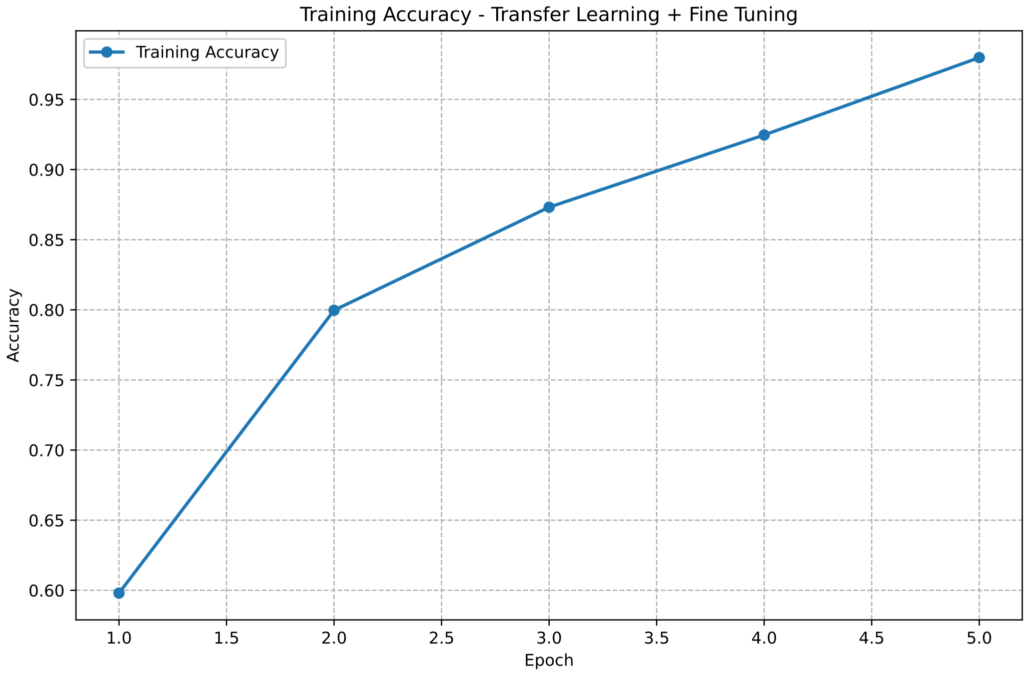
18. Mandatory Plots

1. Sample CIFAR-10 images



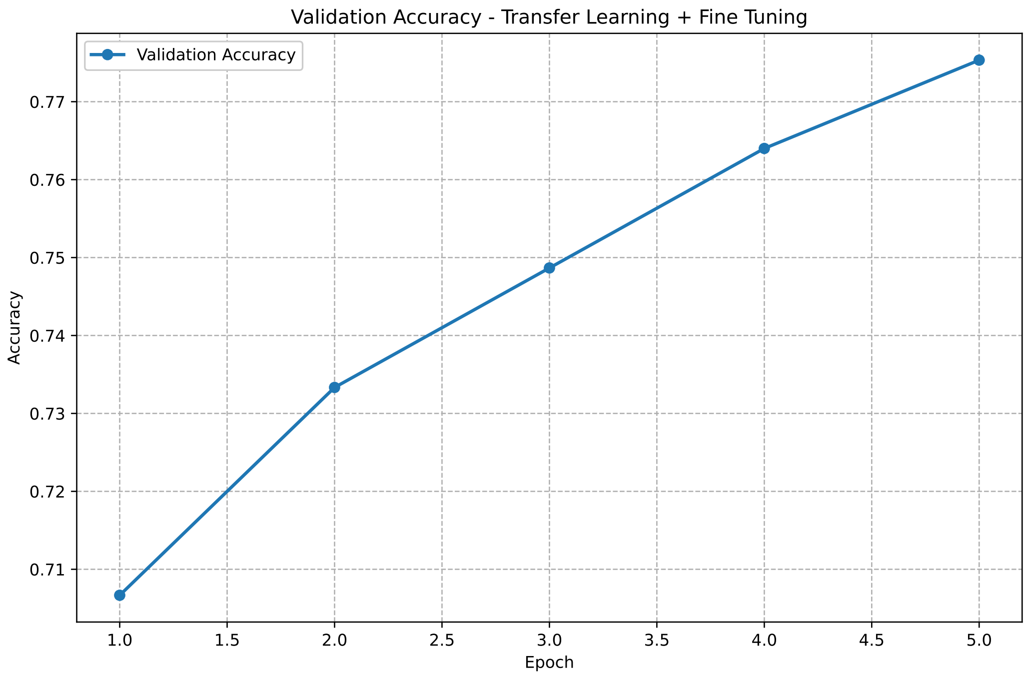
The figure shows representative images from the CIFAR-10 dataset belonging to different classes such as airplane, automobile, bird, cat, and dog. The images demonstrate the diversity of objects used for training and evaluating the VGG16 transfer-learning model.

2. Training Accuracy



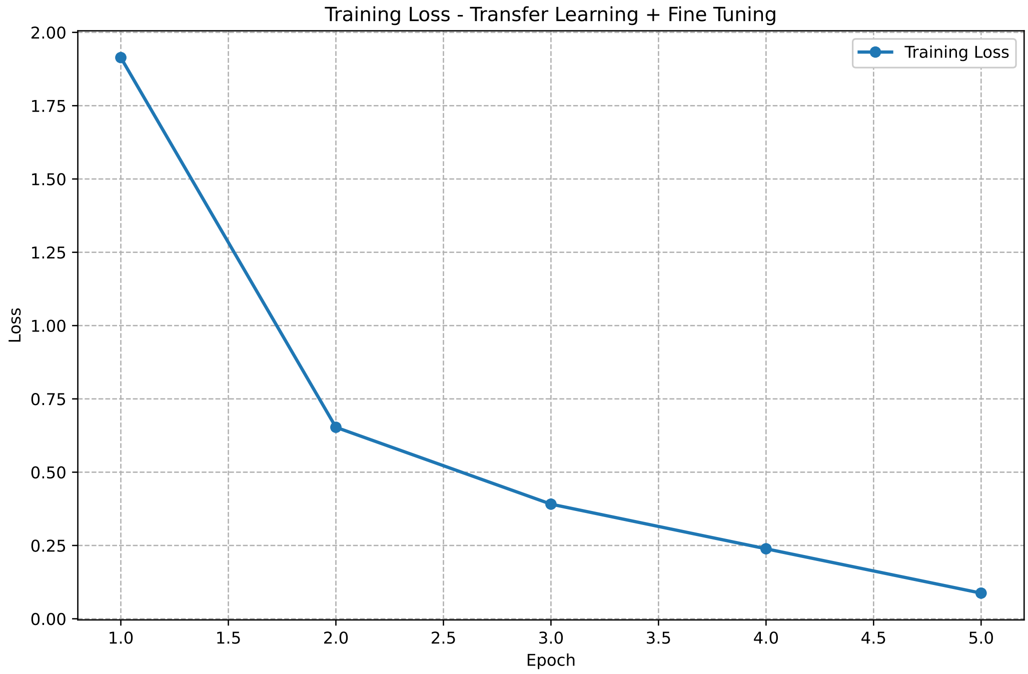
Training accuracy increases progressively across the epochs, indicating that the VGG16 model learns useful features from the CIFAR-10 training data. The final training accuracy is 97.98%.

3. Validation Accuracy



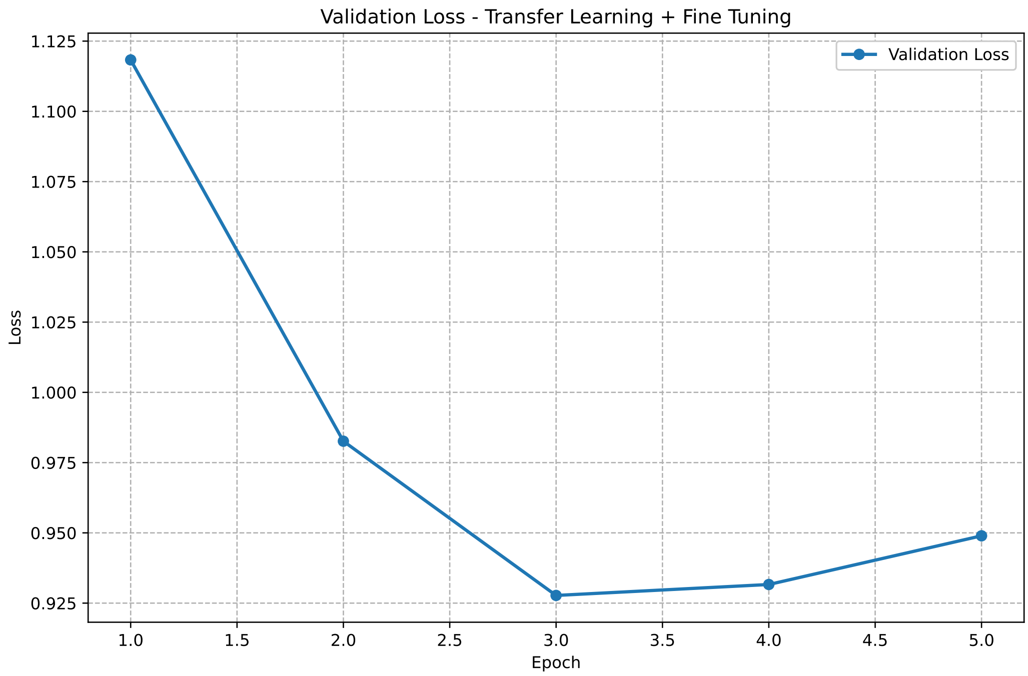
Validation accuracy improves during training, showing that the learned features generalize to unseen validation images. The difference between the high training accuracy and validation accuracy indicates some degree of overfitting.

4. Training Loss



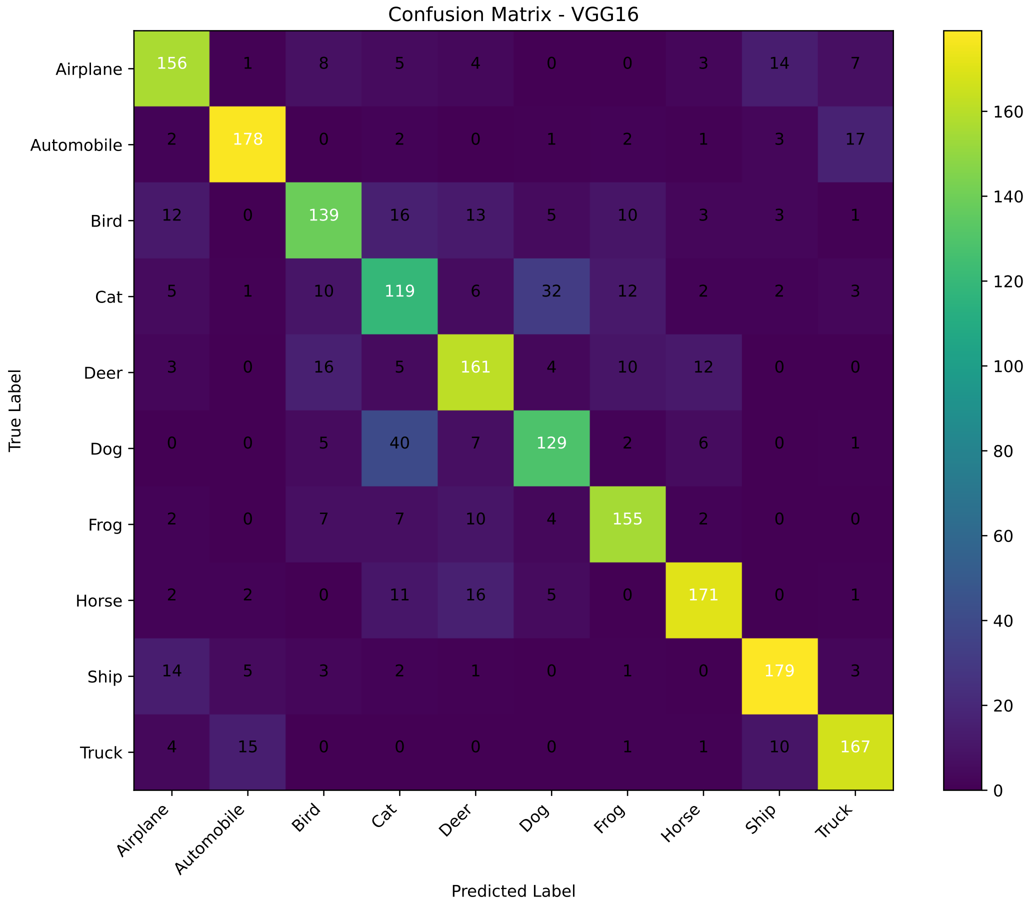
Training loss decreases as the number of epochs increases, indicating that the model's prediction error is reducing during training. The decreasing loss confirms that the model is effectively learning the CIFAR-10 training patterns.

5. Validation Loss



Validation loss generally decreases as the model learns, indicating improved performance on unseen data. If the validation loss begins to increase while training loss continues to decrease, it indicates overfitting to the training dataset.

6. Confusion Matrix



The confusion matrix shows the distribution of correct and incorrect predictions across the ten CIFAR-10 classes. Classes such as Automobile, Frog, Horse, Ship, and Truck show relatively strong classification performance, while Cat and Dog have more confusion with other classes.

7. Misclassified Images (Optional)

The misclassified images show examples where the VGG16 model predicts a class different from the actual class. These errors are expected because CIFAR-10 images have a low resolution of 32×32 pixels, making visually similar classes more difficult to distinguish.

8. Training Time Comparsion

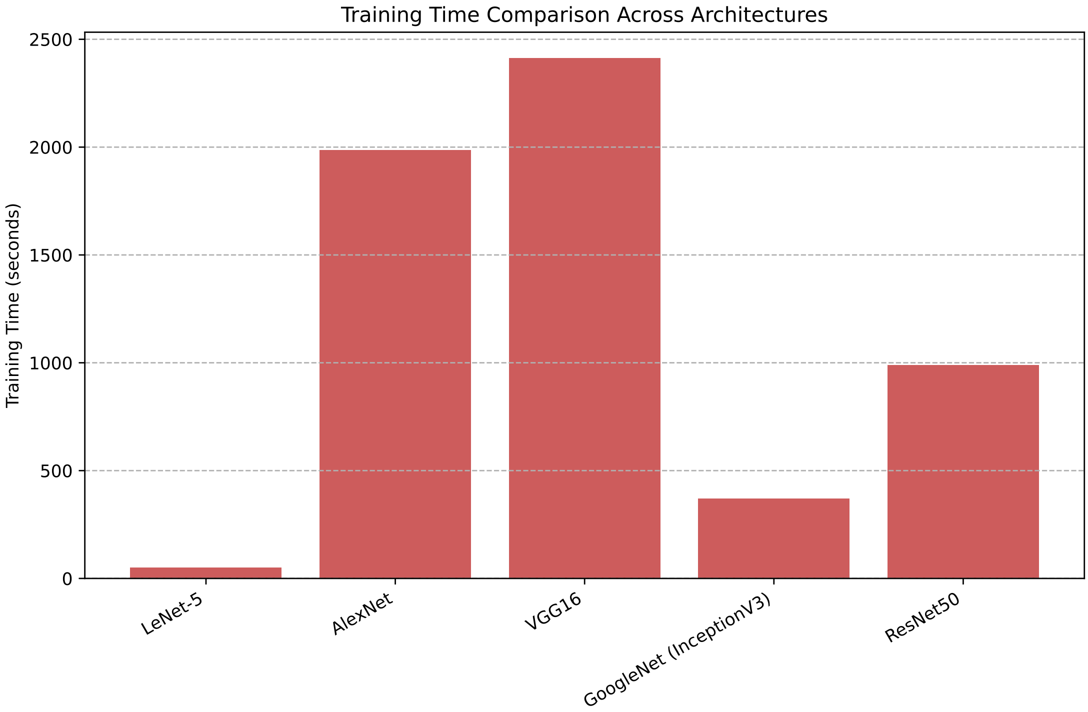


Figure 3: Training Time Comparison of CNN Architectures

ResNet50 achieved a good balance between training time and performance, taking 989.02 seconds. VGG16 required the highest training time of 2412.28 seconds, while LeNet-5 was the fastest at 50.82 seconds. GoogleNet trained considerably faster than AlexNet and VGG16, with a training time of 369.05 seconds.

9. Accuracy Comparsion

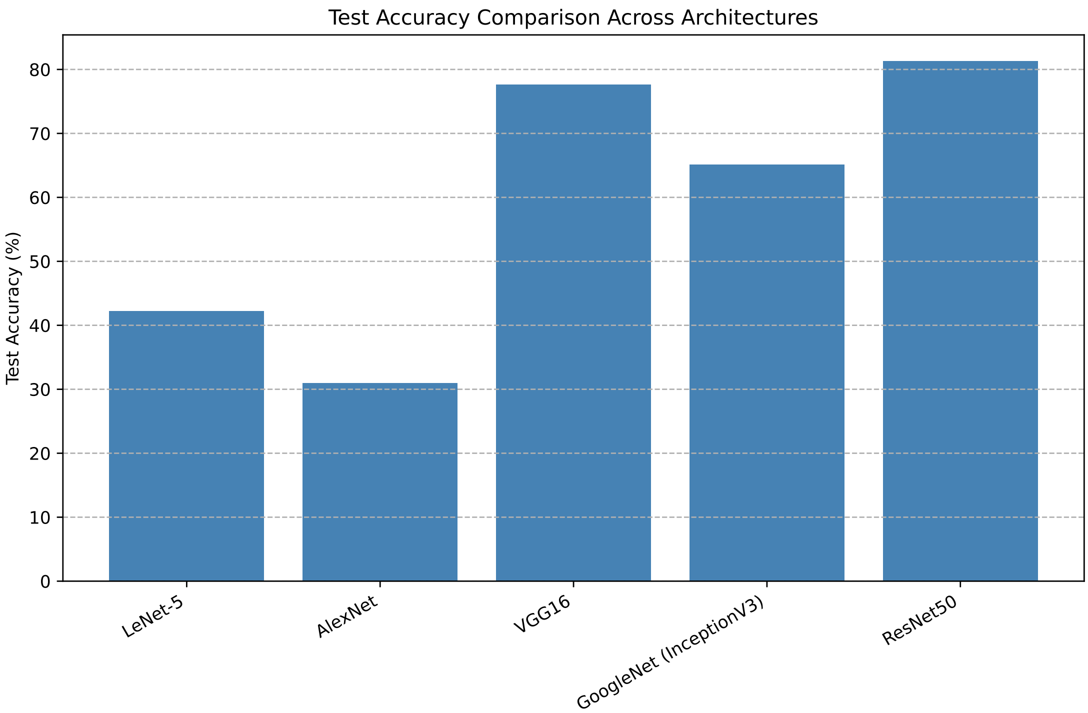


Figure 4: Accuracy Comparison of CNN Architectures

ResNet50 achieved the highest accuracy of 81.33GoogleNet achieved a moderate accuracy of

65.13Overall, ResNet50 provided the best classification performance among the architectures evaluated.

19. Results

19.1 Performance Metrics

Metric	**Value**
Training Accuracy	0.9798
Testing Accuracy	0.7770
Precision	0.7783
Recall	0.7770
F1-score	0.7773
Training Time	4428.18 seconds
Total Parameters	14,781,642

19.2 Comparison of CNN Architectures

Model	**Parameters**	**Accuracy (%)**	**Training Time (s)**
LeNet-5	83,126	42.20	50.82
AlexNet	21,557,642	30.93	1985.30
VGG16	14,781,642	77.60	2412.28
GoogleNet (InceptionV3)	22,066,346	65.13	369.05
ResNet50	23,851,274	81.33	989.02

20. Discussion Questions

1. **Why is AlexNet considered a breakthrough in deep learning?**
AlexNet was a major breakthrough because it showed that a deep convolutional neural network could achieve very high accuracy on large-scale image classification tasks. It made effective use of GPUs for faster training and introduced techniques such as ReLU activation and dropout. Its success on the ImageNet dataset played an important role in bringing deep learning into mainstream computer vision.

2. **Why does VGG16 use only 3×3 convolution filters?**
VGG16 mainly uses small 3×3 filters because they can capture useful local features while keeping the number of parameters relatively low. Multiple 3×3 convolution layers can also provide a larger effective receptive field while adding more nonlinear activation layers. This allows the network to learn complex features without using very large filters.

3. **Explain the advantages of the Inception module.**
The Inception module allows the network to process an image using different filter sizes at the same time. This helps the model capture both small and large patterns present in an image. It also uses 1×1 convolutions to reduce the number of channels, which helps control the computational cost and makes the network more efficient.

4. **What is the purpose of residual learning?**
Residual learning was introduced to make it easier to train very deep neural networks. Instead of directly learning the complete mapping, the network learns the difference, or residual, between the input and the desired output. Skip connections allow the input to bypass some layers, which helps reduce the vanishing gradient problem and makes deeper networks easier to optimize.

5. **Differentiate LeNet and ResNet.**
LeNet is an early and relatively simple CNN designed mainly for handwritten digit recognition. It contains only a few convolutional and pooling layers and has a small number of parameters. ResNet is much deeper and uses residual or skip connections, allowing it to train effectively even with many layers. Therefore, ResNet is more suitable for complex and large-scale image classification tasks.

6. **What is Transfer Learning?**
Transfer learning is the process of using a model that has already been trained on a large dataset and adapting it to a new task. Instead of training a CNN completely from scratch, the pretrained model can be used as a starting point. This is especially useful when the new dataset is relatively small, as the model already contains useful features such as edges, shapes, and textures.

7. **Why is fine tuning required?**
Fine tuning is used to adapt a pretrained model more closely to the new dataset and task. The pretrained layers already contain general image features, but the new dataset may have different characteristics. By training some of the later layers, and sometimes the earlier layers with a smaller learning rate, the model can learn features that are more specific to the new problem.

8. **Explain the difference between dilated convolution and transpose convolution.**
Dilated convolution increases the receptive field by inserting gaps between the elements of a convolution filter without increasing the number of filter parameters significantly. It is useful when we want to capture information from a larger area of an image while maintaining the spatial resolution. Transpose convolution, on the other hand, is mainly used to increase the spatial dimensions of feature maps and is commonly used in image generation and segmentation tasks.

9. **Why do pretrained models converge faster?**
Pretrained models converge faster because they do not start with completely random weights. They have already learned useful visual features such as edges, textures, shapes, and patterns from a large dataset. Therefore, when the model is trained on a new dataset, it only needs to adjust these existing features rather than learning everything from the beginning.

10. **Compare the computational complexity of LeNet and ResNet.**
LeNet has a relatively small number of layers and parameters, so it requires much less computation and memory during training and inference. ResNet is significantly deeper and contains millions of parameters, making it computationally more expensive. However, the residual connections in ResNet make it possible to train such deep networks effectively and achieve much better performance on complex image classification tasks.

21. Additional Exercises

1. **Implement transfer learning using VGG16.**
VGG16 pretrained on ImageNet was used as the base model. Initially, its convolutional layers were frozen and a new classification head was added for the 10 CIFAR-10 classes. After initial training, the fifth convolutional block was unfrozen and fine-tuned using a smaller learning rate.

2. **Repeat the experiment using ResNet50.**
   ResNet50 pretrained on ImageNet was used with a new classification head for CIFAR-10. The pretrained layers were initially frozen. The last residual block was then unfrozen and fine-tuned using a smaller learning rate.
3. **Compare Adam and SGD optimizers.**
   Adam uses adaptive learning rates and generally converges faster, while SGD uses a fixed learning rate and may require careful tuning. Adam was used in this experiment for efficient training and fine-tuning.
4. **Compare training with frozen layers and fine tuning.**
   With frozen layers, only the newly added classification layers are trained, making training faster and reducing overfitting. Fine-tuning unfreezes some pretrained layers, allowing the model to adapt better to the target dataset but requiring more computation.
5. **Evaluate the model using Precision, Recall and F1-score.**
   Precision measures the correctness of positive predictions, while Recall measures how many actual samples are correctly identified. F1-score is the harmonic mean of Precision and Recall.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

   Weighted averaging was used for the CIFAR-10 multiclass evaluation.
6. **Compare the performance of LeNet, AlexNet and ResNet.**
   LeNet is the simplest and least computationally demanding model. AlexNet is deeper and can learn more complex features. ResNet50 is much deeper and uses residual connections along with ImageNet transfer learning. Therefore, ResNet50 provides the best performance among the three.

22. References

1. Yann LeCun, Leon Bottou, Yoshua Bengio and Patrick Haffner, *Gradient-Based Learning Applied to Document Recognition*, Proceedings of the IEEE, 1998.
2. Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton, *ImageNet Classification with Deep Convolutional Neural Networks*, NeurIPS, 2012.
3. Karen Simonyan and Andrew Zisserman, *Very Deep Convolutional Networks for Large-Scale Image Recognition*, ICLR, 2015.
4. Christian Szegedy et al., *Going Deeper with Convolutions*, CVPR, 2015.
5. Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
7. TensorFlow Documentation: `https://www.tensorflow.org`
8. Keras Documentation: `https://keras.io`